

CS 6959 — Project 4

Modifying Hardware Configuration

All configurations use the same kernel and Sponza scene at 512×512 , 1 sample per pixel, up to three ray segments, and a 1.515 GHz simulated clock.

Configuration	Cycles	Frame time (ms)
A: 8 TMs, two-level cache	2,486,319	1.641135
B: 64 TMs, two-level cache	521,272	0.344074
C: 64 TMs, three-level cache	2,302,386	1.519727
Optimized C: 384 MSHRs per L2	1,223,485	0.807581

a. Increasing TMs from 8 to 64

B is **4.77× faster** than A. Scaling is less than $8\times$ because more TMs do not proportionally increase shared memory-system resources or eliminate memory stalls.

b. Adding a third cache level

C takes **4.42× as long** as B. Its group-local L2 caches have a lower hit rate, so many requests must traverse the additional 480-cycle L3 stage. The added hierarchy therefore hurts performance on this workload.

c. L2 hit and miss rates

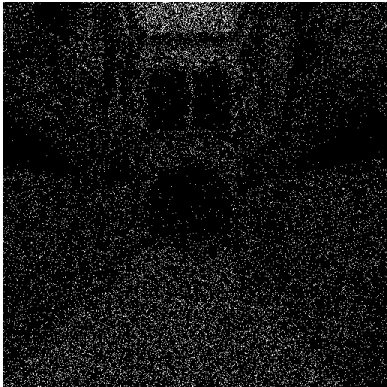
L2 organization	Hit	Half miss	Miss
B: shared across all 64 TMs	80.28%	0.41%	19.31%
C: one L2 per 8 TMs	39.47%	0.36%	60.17%

Half misses merge into an outstanding miss; they are not ordinary hits. Both designs have 4 MiB of L2 in total, but each group in C can use only its own 512 KiB. Restricted sharing and duplicated data across groups help explain the lower hit rate.

d. Selected optimization

Increase only the new L2's `num_mshr` from **192 to 384**. Keep C's topology, 4 MiB L3, 480-cycle L3 latency, and all other hardware parameters unchanged. More MSHRs allow more cache misses to remain in flight, reducing backpressure while data returns.

The change gives **1.88× speedup over C**, reducing frame time by **46.86%**. L2's reported MSHR/queue-stall count falls from 57,419,578 to 23,603,232, while DRAM reads remain approximately 68.3 MB. This supports increased miss concurrency, rather than reduced memory traffic, as the source of improvement.



Sponza, 512×512 . All four configurations produce identical images.